



Computer Workshop Practice

Ground to Mastery — Complete Study Notes

ICT 121 | Year 1 Semester 2 | DCS | UICT Uganda

Tools · Safety · Electronics · Printers · Power Packs · E-Waste

Coursework 40% (Test 10% + Assignment 10% + Practical 20%) | Final Exam 60%

Bravehart · TheRealistDevLab · 2025/2026

How to Use This Guide

ICT 121 is a PRACTICAL course. Theory explains why you do things in the lab. Study by understanding the reasons behind each procedure, not just memorising steps. Each unit has three levels:

BEGINNER

BEGINNER — Understand what each tool, component, or concept IS and what it does.

INTERMEDIATE

INTERMEDIATE — Understand HOW things work, procedures step by step, how to identify problems.

MASTERY

MASTERY — Troubleshoot problems, apply knowledge in real lab situations, answer structured exam questions.

NOTE

Assessment: Test=10% | Assignment=10% | Practical(lab)=20% | Final Exam=60% The practical component (20%) means you WILL be assessed performing real tasks in the lab.

UNIT 1: Laboratory Tools & Safety

A computer workshop uses specialised tools for assembly, maintenance, diagnosis, and repair. Knowing the correct tool for each task prevents damage to equipment and injury to the technician.

BEGINNER

A tool is any instrument or equipment used to perform a particular activity. Tools are grouped into: hand tools, ESD protection, diagnostic tools, cleaning tools, networking tools, and support equipment.

1.1 — Hand Tools

Term	Definition
Flathead screwdriver	Turns slotted (minus) screws — various sizes needed — do not use wrong size
Phillips screwdriver	Turns cross-head (plus) screws — most common type in computers — most used tool
Torx screwdriver	Star-shaped head — used in newer laptops, Macs, and some hard drives
Needle-nose pliers	Grips small components, picks up jumpers, holds small screws in tight spaces
Wire cutters/strippers	Cuts and strips network or power cables — used in cable installation and repair
Tweezers	Handles very small electronic parts that fingers cannot grip safely

1.2 — ESD Protection Equipment

Term	Definition
ESD wrist strap	Anti-static strap worn on wrist — clips to grounded metal — continuously drains static to ground. THE MOST IMPORTANT TOOL.
Anti-static mat	Grounded work surface — dissipates static — connect wrist strap to mat for full protection
Anti-static bag	Pink/silver metallic bags — ALWAYS store components in these when not installed in a system
ESD-safe clothing	Special conductive work clothing — used in professional clean rooms

1.3 — Diagnostic Tools

Term	Definition
Digital Multimeter (DMM)	Measures voltage, current, and resistance — tests PSU voltages, cable continuity, component values

Term	Definition
POST diagnostic card	Plugs into PCIe slot — shows numeric error codes to identify startup failures without monitor
LAN cable tester	Tests all 8 wires in a network cable for correct connections and continuity
Oscilloscope	Displays electrical signals as waveforms — analyses signal timing and quality

1.4 — Cleaning & Network Tools

Term	Definition
Compressed air can	Blows dust from fans, heatsinks, vents — NEVER use your mouth (moisture damages components)
Bristle brush	Removes stubborn dust from heatsink fins and fan blades
Crimping tool	Attaches RJ-45 connectors to Ethernet cables — essential for cable making
Punch-down tool	Connects wires to patch panels or network jacks in structured cabling installations

1.5 — Support Equipment

Term	Definition
Soldering iron and solder	Joins wires and components to circuit boards — repairs electronic circuits
Desoldering pump	Removes existing solder from boards when replacing components
Magnifying lamp	Helps view small components and fine solder joints clearly
Thermal paste	Heat-conducting compound between CPU and heatsink — fills microscopic air gaps
Parts organiser tray	Holds screws and small parts during assembly — prevents loss
Tool kit box	Organises and stores all maintenance tools securely

Computer Workshop Tool Categories

COMPUTER WORKSHOP TOOLS		
HAND TOOLS Screwdrivers Pliers Tweezers	DIAGNOSTIC Multimeter POST card Oscilloscope	ESD PROTECTION Wrist strap Anti-static mat Anti-static bags
CLEANING Compressed air	NETWORKING Crimping Punch-down	SUPPORT EQUIPMENT Soldering iron Desoldering pump

Bristle	LAN tester	Magnifying lamp
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1.6 — Safety Rules

- Always wear ESD wrist strap when handling internal components — static kills chips
- Power OFF and UNPLUG before opening case — never work on live equipment
- Hold circuit boards by edges only — never touch chips, solder, or connectors
- Keep all liquids away from equipment — drops cause short circuits
- Use correct screwdriver size — wrong size strips screw heads permanently
- Never force connectors — check orientation first
- Never work alone — always have someone nearby for emergencies
- No food or drink in the workshop at any time
- Report faulty equipment to workshop supervisor immediately

Electrical Safety

Domestic supply = 240V AC in Uganda — serious electric shock hazard. CRT monitors store up to 30,000 volts even when switched off for some time.

- Always replace blown fuses with identical ampere rating
- Never work alone — keep someone nearby who can assist in emergencies

1.7 — First Aid

Term	Definition
Electric shock	Do NOT touch victim while connected to power — disconnect power FIRST — call for help — CPR if not breathing
Burns	Running cold water for 10+ minutes — NO ice, butter, or toothpaste — seek medical help
Eye injury	Flush with clean water 15+ minutes — do not rub — seek medical attention immediately
Cuts	Firm pressure with clean cloth — elevate limb — seek help for deep cuts
Fire	Use correct extinguisher — evacuate if large — do not re-enter burning building

1.8 — Fire Extinguisher Types

Term	Definition
Class A (Water/Foam)	Wood, paper, textiles — NEVER on electrical fires — water conducts electricity
Class B	Flammable liquids — petrol, oil, solvents
Class C (CO2/Dry powder)	ELECTRICAL fires — computers — CO2 leaves no residue and does not damage equipment

Term	Definition
Class D	Metal fires — magnesium, sodium
CO2 extinguisher	BEST for computer rooms — clean, non-conductive, no equipment damage

⚠️ **WARNING**

NEVER use water or foam on electrical fires. Water conducts electricity and causes electrocution. CO2 is the ONLY safe choice for computer equipment fires.

💡 **EXAM TIP**

Exam: Know 5+ tools and functions. Know all fire extinguisher classes. Know first aid (disconnect power FIRST for electric shock). These appear in MCQ and structured questions every year.

UNIT 2: Computer System Usage & Accessories

This unit covers correct connection procedures, maintenance tasks, and energy configuration. These are practical skills assessed in the lab.

2.1 — Connecting a Computer System

1. Place monitor at comfortable eye level distance from user
2. Connect video cable (VGA, HDMI, or DisplayPort) from monitor to PC
3. Connect keyboard and mouse to USB (or PS/2) ports
4. Connect network cable (RJ-45 Ethernet) to NIC port at back of PC
5. Connect power cables — check voltage selector (240V in Uganda)
6. Power on MONITOR first, then press PC power button

NOTE

Always verify voltage selector is set to 220-240V before connecting power. Wrong voltage destroys the PSU instantly.

2.2 — Basic Maintenance Tasks

Term	Definition
Antivirus scan	Run regularly — ALWAYS update virus definitions before scanning
Disk cleanup	Remove temp files, empty Recycle Bin, clear browser cache — frees disk space
Software updates	Apply OS and app security updates — prevents vulnerabilities
Defragmentation	HDDs ONLY — run monthly — NEVER defragment SSDs
Backup	Regular backups — follow 3-2-1 rule: 3 copies, 2 media types, 1 offsite
Dust removal	Every 3-6 months — compressed air on fans, heatsinks, vents

2.3 — Types of Maintenance

Term	Definition
Preventive	Done BEFORE failures — regular cleaning, updates, inspections — cheapest long-term
Corrective	Done AFTER failure — finding cause, replacing parts, repairs — most expensive
Predictive	Monitors for warning signs using diagnostic tools — CPU temp, SMART data, error logs

2.4 — Computer Maintenance Plan Steps

- Checking computer log files for errors and security alerts
- Assessing hard disk space — ensure adequate free space
- Monitoring computer temperature — prevent thermal damage
- Examining security features — antivirus, firewall, updates
- Installing security patches and critical software updates
- Performing regular comprehensive backups of all important data

2.5 — Energy Star Configuration

- Enable sleep mode after inactivity — reduces power consumption
- Set monitor to turn off after 10-15 minutes idle
- Enable power management in BIOS/UEFI
- Unplug unused devices — standby still draws phantom power

2.6 — Built-in Windows Diagnostic Tools

Term	Definition
perfmon.msc	Performance Monitor — tracks CPU, RAM, disk, network — identifies bottlenecks
resmon	Resource Monitor — real-time CPU, memory, disk, network with process details
mdsched	Windows Memory Diagnostic — tests RAM for errors on next restart
CHKDSK	Scans and repairs file system errors — run: <code>chkdsk C: /f</code>
dxdiag	DirectX Diagnostic — checks graphics, sound, DirectX components

💡 EXAM TIP

Know all 3 maintenance types. Preventive vs Corrective vs Predictive. Know the 6 basic maintenance tasks. Know Energy Star settings. These appear in EVERY exam.

UNIT 3: Electronics Fundamentals for Maintenance

Understanding basic electronics is essential for maintenance. Technicians must identify components, read colour codes, and test with a multimeter.

BEGINNER

BEGINNER: Electronic components are the building blocks of all circuits. Classified as PASSIVE (no external power needed) or ACTIVE (require external power). Computers are made of thousands of these components.

3.1 — Passive vs Active Components

Passive Components

Term	Definition
Characteristics	No external power needed. Cannot amplify. Consume, store, or dissipate energy. Bidirectional.
Resistor (R)	Opposes current flow — dissipates heat. Ohm's Law: $V=IR$. Unit: Ohms (Ω).
Capacitor (C)	Stores energy in electric field — blocks DC, passes AC. $Q=C \times V$. Unit: Farads (F).
Inductor (L)	Stores energy in magnetic field — opposes current change. $V=L(dI/dt)$. Unit: Henry (H).
Transformer	Transfers energy magnetically — no power gain — fundamental to power grids
Fuse	Sacrificial protection — melts when current exceeds rating — breaks circuit

Active Components

Term	Definition
Characteristics	Require external power. Can amplify, switch, or generate signals. Usually unidirectional.
Diode	One-way current valve — rectification. LEDs emit light using energy.
Transistor (BJT/MOSFET)	Amplifies or switches signals — core of all modern electronics
Integrated Circuit (IC)	Contains many transistors — CPU, microcontrollers, logic gates, op-amps

3.2 — Capacitors in Detail

A capacitor stores electrical energy in an electric field between two conducting plates separated by a dielectric (insulating material). Also called a condenser.

- Stores charge: $Q = C \times V$ (Coulombs = Farads $\times$ Volts)
- In DC: charges to supply voltage then BLOCKS further current flow

- In AC: current appears to pass through freely — used for signal coupling
- Applications: power supply smoothing, audio coupling, RF filters, tuned circuits

3.3 — Inductors in Detail

An inductor stores energy in a magnetic field. Also called a coil, choke, or reactor. Made of insulated wire wound around a core.

- Works on Faraday's Law of Electromagnetic Induction
- BLOCKS AC while allowing DC to pass (opposite to capacitor)
- Applications: power supply ripple removal, radio tuning circuits, transformers

3.4 — Electrostatic Discharge (ESD)

Static builds up on your body from movement. When you touch a component, it discharges — potentially destroying it. Just 20V can destroy a chip running on 3.3V or 5V. You feel NOTHING, but the chip is dead.

Term	Definition
ESD wrist strap	Continuously drains static to ground — MOST IMPORTANT TOOL — always worn during internal work
Anti-static bag	Store ALL components in these when not installed — never on regular plastic surfaces
Anti-static mat	Grounded work surface — connect strap to mat for complete protection
Humidity	Low humidity = more static. Work in humid environment when possible.

3.5 — Resistor Colour Code

Resistors have coloured bands indicating value, multiplier, and tolerance. IEC 60062/EIA standard. Read LEFT to RIGHT — tolerance band (Gold/Silver) is usually on the right and wider-spaced.

Colour Table

Colour	Digit
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7

Colour	Digit
Grey	8
White	9
Gold (tolerance)	±5%
Silver (tolerance)	±10%

REMEMBER

Memory Mnemonic: BB ROY Great Britain Very Good Wife Black Brown Red Orange Yellow Green Blue Violet Grey White = 0 1 2 3 4 5 6 7 8 9

Band Formats

- 4-band (most common): Digit1 — Digit2 — Multiplier — Tolerance
- 5-band (precision): Digit1 — Digit2 — Digit3 — Multiplier — Tolerance
- 6-band (rare): adds Temperature Coefficient (TCR) as 6th band

Reading Resistors — Examples

Brown-Black-Red-Gold:

Brown=1, Black=0, Red= $\times 100$, Gold= $\pm 5\%$

$\rightarrow 10 \times 100 = 1,000\Omega = 1k\Omega \pm 5\%$ (range: 950Ω to $1,050\Omega$)

Red-Violet-Orange-Gold:

2, 7, $\times 1000$, $\pm 5\%$ $\rightarrow 27k\Omega \pm 5\%$

Yellow-Violet-Green-Silver:

4, 7, $\times 100k$, $\pm 10\%$ $\rightarrow 4.7M\Omega \pm 10\%$

3.6 — Capacitor & Inductor Colour Coding

Modern capacitors use printed numbers (e.g. "104" = 100nF). Older types use colour bands — value in picofarads (pF). Same colours as resistors.

- Example: Brown-Black-Orange = $10 \times 1,000 = 10,000 \text{ pF} = 10 \text{ nF}$

Inductors (axial types): same system as resistors — value in microhenries (μH).

- Example: Red-Violet-Gold = $27 \times 0.1 = 2.7 \mu\text{H}$

3.7 — Using a Digital Multimeter (DMM)

A DMM measures voltage (V), current (A), and resistance (Ω). Most essential diagnostic tool. Face has: Display, Buttons, Rotary Dial, Input Jacks. Red=positive, Black=negative/COM.

Term	Definition
Voltage DC (V $\rightarrow$)	Probes ACROSS component — black to negative, red to positive — set range above expected
Voltage AC (V $\sim$)	Same placement — for mains/AC power testing — set to AC range
Current (A)	Meter in SERIES — break circuit, insert meter — red to A/mA

Term	Definition
	terminal
Resistance (Ω)	POWER OFF first — probes across component — set Ω range
Continuity	POWER OFF — beep = complete circuit — tests cables, fuses, switches
Diode test	~0.6V for good silicon diode — 0 = short circuit — OL = open

Measuring Voltage — Step by Step

7. Plug black lead into COM, red into V terminal
8. Set dial to V $\overline{=}$ (DC) or V $\sim$ (AC) as required
9. Set range HIGHER than expected voltage
10. Touch red probe to positive terminal, black to negative
11. Read display: AA battery ~1.5V, 9V battery ~9V, mains ~240V AC

Measuring Resistance — Step by Step

12. POWER OFF the component — always — live circuits damage meter
13. Insert black lead into COM, red into Ω terminal
14. Set dial to Ω range — start higher than expected
15. Touch probes across both ends of component
16. Compare reading to colour code — should be within tolerance

3.8 — Testing Basic Components

Term	Definition
Fuse	Continuity — BEEP=good, No beep=blown. Replace same ampere rating.
Switch	Continuity — BEEP when ON, silent when OFF
Cable/wire	Continuity — beep=intact, no beep=broken somewhere
Battery	DC voltage — AA~1.5V, 9V~9V, 12V car~12.6V charged
Resistor	Resistance (Ω) — compare to colour code value — within tolerance=good
Bulb/lamp	Continuity — beep=filament intact, no beep=blown
PSU	Paperclip test: short green to black → fan spins = PSU alive



PRACTICAL

Lab Experiment: Resistor Reading and Verification

1. Pick three different resistors from the lab set
2. Read colour code — record theoretical value and tolerance
3. Calculate tolerance range (e.g. 1k Ω $\pm$ 5% = 950 Ω to 1,050 Ω)
4. Set DMM to Ω , power off, probes on resistor leads
5. Record measured value — is it within tolerance range?
6. Record in lab table: Resistor | Colour Code | Theoretical | Measured | Within tolerance?

 EXAM TIP

Colour code MUST be memorised. Know the mnemonic. Know 4-band and 5-band reading. Know DMM settings for each measurement. The practical is 20% of your grade — practise with real components.

UNIT 4: Printers, Scanners & Photocopiers

A printer is an output device producing text/graphics on paper (hard copy). Printers are IMPACT (strike mechanism hits ribbon/paper) or NON-IMPACT (no physical contact with paper).

BEGINNER

BEGINNER: Impact = hits paper like a typewriter (noisy, good for receipts/forms). Non-impact = quiet, modern, higher quality. Speed = ppm (pages per minute) or cps (chars/sec). Quality = dpi (dots per inch).

4.1 — Types of Printers

Term	Definition
Daisy Wheel	IMPACT — rotating wheel — hammer strikes spoke — letter quality — CANNOT print graphics — very slow
Dot-Matrix	IMPACT — pins strike ribbon — 9-pin or 24-pin (24 better) — prints multipart forms — noisy — dusty environment OK
Line Printer	IMPACT — entire line at once — very fast — used with mainframes — speed in lpm
Inkjet	NON-IMPACT — CMYK ink droplets through nozzles — colour — quiet — expensive ink — may smear
Laser	NON-IMPACT — toner + heat + drum — fastest — 600-1200 dpi — cheapest per page for high volume
Thermal	NON-IMPACT — heat-sensitive paper — no ink/toner — receipts, labels — quiet
3D Printer	NON-IMPACT — builds objects layer by layer — plastic filament or resin
Plotter	NON-IMPACT — moves pen — engineering drawings — precise lines

4.2 — How a Laser Printer Works

Also called a page printer — processes and stores the ENTIRE page before printing begins.

Laser Printer Process

1. CHARGING → Drum given uniform negative charge
2. EXPOSING → Laser removes charge from image areas on drum
3. DEVELOPING → Toner (dry powder) attracted to exposed areas
4. TRANSFERRING → Toner moves from drum to paper
5. FUSING → Heat (180°C) + pressure permanently bonds toner to paper
6. CLEANING → Drum cleaned, discharged, ready for next page

Memory aid: Charge → Expose → Develop → Transfer → FUSE → Clean

4.3 — How an Inkjet Printer Works

17. Print head moves across page on a carriage rail

18. Hundreds of tiny nozzles fire precise ink droplets onto paper
19. CMYK colours (Cyan, Magenta, Yellow, Black) applied in multiple passes
20. Paper advances between passes until page complete

4.4 — Printer Types Comparison

Factor	Details
Speed	Line printer fastest. Laser 4-30+ ppm. Inkjet slower. Dot-matrix slowest (cps).
Quality	Laser 600-1200 dpi. Inkjet 300-4800 dpi. Dot-matrix lowest.
Cost per page	Laser cheapest for high volume. Inkjet expensive ink cartridges.
Multipart forms	Dot-matrix ONLY — must physically impact paper to print through multiple layers
Noise	Dot-matrix noisiest. Laser moderate. Inkjet quiet. Thermal silent.

4.5 — Printer Connectivity

Term	Definition
USB	Most common — direct connection — plug and play — single PC only
Network (Ethernet)	RJ-45 cable — shared across LAN — assign static IP for reliability
WiFi	Wireless — shared across network — convenient for home/office
Bluetooth	Short range — mobile printing from phones
Parallel port (LPT)	Old 25-pin — obsolete — not in modern systems

4.6 — Printer Problems & Troubleshooting

Term	Definition
No output	Check power, cable/network, driver installed, clear print queue of stuck jobs
Poor print quality	Low ink/toner, dirty print head, wrong paper — run head cleaning cycle
Paper jam	Open ALL doors — remove paper GENTLY without tearing — check rollers
Streaks on output	Dirty drum (laser) or clogged nozzle (inkjet) — clean or replace
Printer offline	Check connection, restart Print Spooler, verify network IP address
Slow printing	Draft mode enabled, network congestion, insufficient printer memory

4.7 — Scanner Types & Operation

Term	Definition
Flatbed scanner	Glass surface with lid — highest quality — photos, books, bound documents
Sheet-fed scanner	Document auto-fed through — faster — CANNOT scan bound documents
Handheld scanner	Portable — user drags manually — lower quality
Drum scanner	Professional publishing — highest resolution — complex and expensive

Scanner Operation Steps

21. Connect scanner via USB or network — install driver software
22. Power on scanner and wait for warm-up
23. Place document face-down on glass or insert into sheet-feeder
24. Run PRE-SCAN — quick low-resolution preview to check positioning
25. Adjust crop, set resolution (min 300 DPI documents, 600+ for photos)
26. Click SCAN — save in appropriate format (PDF for documents, JPEG/TIFF for photos)

4.8 — Photocopiers

Term	Definition
Multifunctional (MFP)	Printing + scanning + copying + faxing — network connectivity — most common office
Desk printer/copier	Compact — small office use — lower volume
Wide format	Large documents — engineering drawings, posters — A1/A0 sizes
Label printer	Adhesive labels — thermal or inkjet technology

EXAM TIP

Know ALL printer types. Know the laser process (6 steps). Know how inkjet works (CMYK). Know impact vs non-impact. Know scanner DPI (300 min for docs, 600+ for photos). Pre-scan is always first. Structured questions on printers appear frequently.

UNIT 5: Power Packs & UPS

The PSU converts mains electricity to DC voltages. The UPS provides backup power and/or voltage regulation. In Uganda where power is unstable, understanding both is critical.

5.1 — Power Supply Unit (PSU)

Converts 220-240V AC mains to stable DC voltages required by components. Contains transformer, rectifier, capacitors, inductors, voltage regulators, fuses, and a cooling fan.

Term	Definition
Function	Converts 240V AC to +12V, +5V, +3.3V DC for computer components
+12V (yellow wire)	CPU, GPU, motors in HDDs, cooling fans — highest demand
+5V (red wire)	Motherboard logic, USB ports, older drives
+3.3V (orange wire)	RAM, PCIe slots, M.2 slots, modern motherboard components
-12V (blue wire)	Some serial COM ports — rarely used in modern systems
Ground (black wire)	0V reference — all measurements relative to ground
PS_ON (green wire)	Short to ground = PSU turns on — used for paperclip test
Wattage	Total power output — system needs + 20-30% headroom
80 PLUS	Efficiency: Bronze(82%), Silver(85%), Gold(87%), Platinum(90%), Titanium(94%)

PSU Wire Colours Quick Reference

Yellow = +12V → CPU, GPU, drive motors, fans
 Red = +5V → USB, motherboard logic, older drives
 Orange = +3.3V → RAM, PCIe, M.2, modern components
 Blue = -12V → Some serial ports (rarely needed)
 Black = GND → Ground/reference (0V)
 Green = PS_ON → Short to black = PSU powers on

Paperclip Test: Short GREEN to BLACK wire
 Fan spins immediately = PSU is alive
 No fan = PSU has failed

PSU Connectors

Term	Definition
24-pin ATX	Main motherboard power — always required — large flat connector
4/8-pin EPS	Extra CPU power — near CPU socket — computer WILL NOT boot without this
6/8-pin PCIe	Discrete graphics card (GPU) power — required for any dedicated GPU
SATA Power (15-pin)	Powers SATA hard drives, SSDs, and optical drives

Term	Definition
Molex (4-pin)	Older drives, some fans — being phased out in modern systems

AT vs ATX Comparison

Feature	AT vs ATX
MB connector	AT: 2 connectors (P8+P9, 12 pins) ATX: 1 connector (24 pins)
Power switch	AT: Switches mains directly (dangerous) ATX: Controlled by motherboard (safe)
Voltages	AT: +5V,+12V,-5V,-12V ATX: +3.3V,+5V,+12V,-12V,+5Vsb
Wattage	AT: under 250W ATX: 300W to 1000W+
Modern use	AT: Obsolete ATX: Current standard

5.2 — UPS (Uninterruptible Power Supply)

UPS provides battery backup when mains fails AND/OR regulates voltage. Critical in Uganda where power cuts and voltage fluctuations are frequent.

Term	Definition
Offline/Standby	Switches to battery when mains fails — slight delay (ms) — cheapest — basic protection only
Line-interactive	Regulates voltage fluctuations WITHOUT switching to battery — RECOMMENDED FOR UGANDA — good balance
Online/Double-conversion	Always on battery — zero switchover delay — cleanest power — most expensive — for servers
Stabiliser (AVR)	Regulates voltage ONLY — NO battery backup — no protection during actual power cuts
VA rating	Volt-Ampere capacity — must exceed total wattage of connected equipment
Runtime	How long battery lasts — depends on battery size vs load applied

NOTE

UGANDA RECOMMENDATION: Line-interactive UPS. It handles both voltage instability AND power outages at moderate cost. Online UPS is ideal but expensive. Offline UPS does NOT handle voltage fluctuations.

5.3 — Power Problem Troubleshooting

Term	Definition
Won't turn on	Check wall socket → check PSU rear switch ON → paperclip test → check power button header cable

Term	Definition
Random shutdowns	Overheating (clean dust, check fans) → failing PSU (test voltages) → power fluctuations (add UPS)
Beeping on startup	POST codes: 1 short beep=normal. Multiple beeps=hardware error. Count and identify.
PSU failure signs	Won't power on, burning smell, fan not spinning, random reboots
Surge damage	Install surge protector or UPS — never plug directly into Uganda mains without protection



PRACTICAL

PSU Testing with Multimeter: 1. Paperclip test: Short GREEN to BLACK wire — fan MUST spin = PSU alive 2. Set DMM to DC voltage 3. Yellow to Black = should read +12V (acceptable: 11.4V to 12.6V) 4. Red to Black = should read +5V (acceptable: 4.75V to 5.25V) 5. Orange to Black = should read +3.3V (acceptable: 3.13V to 3.46V) Outside these ranges = PSU failing → replace immediately



EXAM TIP

Know ALL wire colours: Yellow=+12V, Red=+5V, Orange=+3.3V, Black=GND, Green=PS_ON. Know 3 UPS types and when to use each. Line-interactive = Uganda recommendation. Know the paperclip test. Know AT vs ATX differences.

UNIT 6: E-Waste Management

E-waste (electronic waste) refers to discarded electrical or electronic devices. It is the fastest-growing waste category globally. Improper disposal causes serious, long-term environmental and health damage.

6.1 — Categories of E-Waste

Term	Definition
Large IT equipment	Servers, mainframes, large printers, photocopiers, network racks
Small IT equipment	PCs, laptops, monitors, printers, scanners, keyboards, mice, hard drives
Telecom equipment	Mobile phones, routers, switches, landlines, modems
Consumer electronics	TVs, radios, cameras, game consoles, stereos, DVD players
Household appliances	Microwaves, washing machines, refrigerators, electric cookers
Lighting equipment	Fluorescent tubes, CFL bulbs, LED strips (contain hazardous materials)

6.2 — Hazardous Materials in E-Waste

Term	Definition
Lead (Pb)	WHERE: CRT monitors, solder, lead-acid batteries. HEALTH: Neurotoxin — brain and nervous system damage — especially harmful to children
Mercury (Hg)	WHERE: LCD backlights, thermostats. HEALTH: Accumulates in food chain — kidney and brain damage — severe threat via contaminated fish
Cadmium (Cd)	WHERE: Rechargeable NiCd batteries, semiconductors. HEALTH: Carcinogen — kidney damage, bone disease
Brominated flame retardants	WHERE: Circuit board plastic casings, cables. HEALTH: Toxic dioxins when burned — lung damage, hormone disruption
Chromium VI	WHERE: Metal coatings. HEALTH: Carcinogen — skin and respiratory damage
Barium (Ba)	WHERE: CRT glass screens. HEALTH: Muscle weakness, heart problems, organ damage

Environmental Effects

- Soil contamination — chemicals leach into ground — poisons farmland and groundwater
- Air pollution — burning e-waste releases toxic gases — severe respiratory damage
- Water pollution — chemicals reach rivers and lakes — kills aquatic life, contaminates drinking water

- Food chain contamination — mercury bioaccumulates in fish — humans affected by eating contaminated fish
- Informal recycling — workers (often children) exposed to toxins without protection

6.3 — E-Waste Management Strategies

Term	Definition
Personal level	REDUCE: buy durable, repairable equipment. REUSE: donate working items to schools/charities. RECYCLE: certified recyclers only — never general rubbish.
Institutional level	IT asset register. Secure data wipe before disposal. Contracts with certified recyclers. Equipment lifecycle policy.
Government level	E-waste legislation. Certified recycling centres. Regulate import of old electronics. Extended Producer Responsibility (EPR).

6.4 — The E-Waste Management Cycle

27. COLLECTION — Designated drop-off points at offices, schools, retailers
28. SORTING — Separate by type: IT equipment, consumer electronics, hazardous items
29. DISASSEMBLY — Take apart devices by hand to separate different materials
30. MATERIAL RECOVERY — Extract metals (gold, copper, aluminium) from boards
31. SAFE DISPOSAL — Hazardous materials (mercury, lead, acids) via licensed facilities

6.5 — Advanced Maintenance Tasks

- Run disk defragmenter — Windows: Start → Accessories → System Tools → Disk Defragmenter (HDD only)
- Create file backups — external drive or cloud (Google Drive, OneDrive)
- Download utility tools and OS/programme updates via internet
- Clean drives using disk cleaning software (CCleaner, built-in Disk Cleanup)
- Run CHKDSK: `chkdsk C: /f /r` — checks and repairs file system errors

REMEMBER

The 3Rs of E-Waste in ORDER of preference: 1. REDUCE — buy less, buy durable — best option 2. REUSE — donate, refurbish working equipment 3. RECYCLE — certified facilities only — last resort Never DUMP e-waste in general rubbish — toxic and often illegal.

EXAM TIP

MUST KNOW: 3 hazardous materials + health effects. E-waste cycle (5 steps). Management at all 3 levels (personal, institutional, government). The 3Rs. These appear in EVERY exam without fail.

SECTION A: Multiple Choice Questions — Practice & Answers

40 Marks | 2 marks per question | Choose the BEST answer

Q1. Which fire extinguisher is BEST for a computer equipment fire?

- A. Class A Water
- B. Class B Foam
- C. CO2 Carbon Dioxide**
- D. Class D Dry Powder

Answer: C. CO2 Carbon Dioxide

Explanation: CO2 is clean, non-conductive, leaves no residue. Water = electrocution. Foam = damages equipment. Never use either on electrical fires.

Q2. An ESD wrist strap works by:

- A. Adding electrical charge to components
- B. Draining static from your body to ground continuously**
- C. Blocking all voltage from reaching components
- D. Making you immune to electric shock

Answer: B. Draining static from your body to ground continuously

Explanation: ESD strap creates a continuous path (through a 1MΩ resistor) from your body to ground — safely draining any static buildup before it can discharge through components.

Q3. The resistor colour code mnemonic BB ROY Great Britain Very Good Wife represents:

- A. Fire extinguisher types
- B. Digits 0 through 9 for colours Black Brown Red Orange Yellow Green Blue Violet Grey White**
- C. Types of computer buses
- D. Types of maintenance

Answer: B. Digits 0 through 9 for colours Black Brown Red Orange Yellow Green Blue Violet Grey White

Explanation: BB=Black(0), Brown(1), ROY=Red(2), Orange(3), Yellow(4), Great=Green(5), Britain=Blue(6), Very=Violet(7), Good=Grey(8), Wife=White(9).

Q4. When measuring resistance with a multimeter, you MUST:

- A. Set to AC voltage range
- B. Keep the component powered on
- C. Power OFF the component first**
- D. Insert red lead into COM

Answer: C. Power OFF the component first

Explanation: ALWAYS power off before measuring resistance. A live circuit gives wrong readings and can permanently damage the multimeter.

Q5. Which printer can print multipart/carbon copy forms?

- A. Laser printer
- B. Inkjet printer
- C. Dot-matrix printer**
- D. Thermal printer

Answer: C. Dot-matrix printer

Explanation: Only impact printers (dot-matrix) can print through multiple layers. They physically strike the paper — non-impact printers spray or use heat and cannot penetrate multiple layers.

Q6. The BEST UPS type for Uganda's frequent voltage fluctuations is:

- A. Offline/Standby UPS
- B. Line-interactive UPS**
- C. Online/Double-conversion UPS
- D. Voltage stabiliser

Answer: B. Line-interactive UPS

Explanation: Line-interactive UPS regulates voltage fluctuations without switching to battery AND provides backup during complete outages. Offline UPS does NOT regulate voltage — only switches to battery.

Q7. Which wire colour is +12V in an ATX power supply?

- A. Red
- B. Orange
- C. Yellow**
- D. Blue

Answer: C. Yellow

Explanation: Yellow=+12V. Red=+5V. Orange=+3.3V. Blue=-12V. Black=Ground. Green=PS_ON. This must be memorised.

Q8. To test a fuse with a multimeter, use:

- A. DC Voltage setting
- B. Resistance (Ω) setting
- C. Continuity (beep) setting**
- D. Current (A) setting

Answer: C. Continuity (beep) setting

Explanation: Continuity: good fuse = beep (complete circuit). Blown fuse = no beep (open circuit). Power must be OFF.

Q9. The correct first step in laser printing after the document is sent is:

- A. Developing
- B. Transferring
- C. Charging**
- D. Fusing

Answer: C. Charging

Explanation: ORDER: Charging (drum charged) → Exposing (laser writes) → Developing (toner added) → Transferring → Fusing → Cleaning.

Q10. Which hazardous e-waste material accumulates in the food chain through fish?

- A. Lead
- B. Mercury**
- C. Cadmium
- D. Barium

Answer: B. Mercury

Explanation: Mercury from LCD backlights enters water — absorbed by fish (bioaccumulation) — multiplies up food chain (biomagnification) — damages kidneys and brain of humans who eat contaminated fish.

Q11. Minimum DPI for scanning a document is:

- A. 72 DPI
- B. 150 DPI
- C. 300 DPI**
- D. 600 DPI

Answer: C. 300 DPI

Explanation: 300 DPI is the minimum for document scanning to maintain readability. 600+ DPI for photos and detailed images where quality matters.

Q12. Preventive maintenance is BEST described as:

- A. Fixing problems after they occur

- B. Monitoring warning signs of failure
- C. Regular tasks done before failures happen**
- D. Updating system after hardware changes

Answer: C. Regular tasks done before failures happen

Explanation: Preventive = done on a SCHEDULE before problems occur — cleaning, updates, backups. Most cost-effective long-term approach.

Q13. A capacitor stores energy in:

- A. A magnetic field
- B. A chemical reaction
- C. An electric field**
- D. A mechanical spring

Answer: C. An electric field

Explanation: Capacitors store in ELECTRIC FIELD between two plates. Inductors store in MAGNETIC FIELD. Batteries store via chemical reactions.

Q14. Which type of printer has the LOWEST print quality?

- A. Inkjet
- B. Laser
- C. Dot-matrix**
- D. Thermal (photo)

Answer: C. Dot-matrix

Explanation: Dot-matrix quality is lowest because pins strike through a ribbon creating relatively coarse dots. Laser and inkjet produce much sharper output.

Q15. The 3Rs of e-waste management in correct priority order are:

- A. Repair, Reuse, Recycle
- B. Reduce, Repair, Remove
- C. Reduce, Reuse, Recycle**
- D. Remove, Replace, Recycle

Answer: C. Reduce, Reuse, Recycle

Explanation: REDUCE (buy less, buy durable) → REUSE (donate, refurbish) → RECYCLE (certified facility). In that order of preference.

Q16. Gold tolerance band on a resistor means:

- A. The resistor is made of gold
- B. Tolerance $\pm 10\%$
- C. Tolerance $\pm 5\%$**
- D. No tolerance — exact value

Answer: C. Tolerance $\pm 5\%$

Explanation: Gold = $\pm 5\%$. Silver = $\pm 10\%$. No band = $\pm 20\%$ (lowest precision). Gold is more precise (closer to stated value) than silver.

Q17. When someone receives an electric shock, your FIRST action should be:

- A. Apply cold water immediately
- B. Give them something to drink
- C. Disconnect the power source safely**
- D. Begin CPR immediately

Answer: C. Disconnect the power source safely

Explanation: NEVER touch victim while in contact with electricity — you will be shocked too. FIRST disconnect the power source, THEN help the victim and call for assistance.

Q18. The paperclip test on a PSU involves:

- A. Inserting a paperclip into the power socket

B. Shorting the green wire to a black wire

C. Measuring resistance with a paperclip

D. Bending pins with a paperclip

Answer: B. Shorting the green wire to a black wire

Explanation: Short green (PS_ON) to any black (GND) wire using a bent paperclip. Fan should spin immediately = PSU is alive. No fan = PSU dead.

Q19. A PRE-SCAN in scanning software means:

A. The final high-resolution scan

B. A quick low-resolution preview before the final scan

C. Scanning a document twice

D. Installing the scanner driver

Answer: B. A quick low-resolution preview before the final scan

Explanation: Pre-scan = quick preview at low resolution. Allows you to check positioning, crop the image, and set resolution before doing the final proper scan.

Q20. Which step comes AFTER transferring in the laser printing process?

A. Charging

B. Exposing

C. Developing

D. Fusing

Answer: D. Fusing

Explanation: ORDER: Charge → Expose → Develop → Transfer → FUSING (heat bonds toner to paper) → Cleaning. Fusing is always AFTER transferring.

SECTION B: Structured Questions — Answer Any THREE

Answer ANY THREE questions. Each question = 20 marks. Show full reasoning.

Question 1

20

Answer all parts of this question.

(a) **10 marks** Name FIVE computer workshop tools and explain the purpose of each.

Model Answer: 1. ESD Wrist Strap: Worn on wrist, clipped to grounded metal. Continuously drains static electricity from the technician's body to ground — prevents ESD from destroying sensitive components like CPUs, RAM, and ICs. 2. Phillips Screwdriver: Used to turn cross-head (plus) screws — the most common screw type in computers. Must use correct size to avoid stripping screw heads. 3. Digital Multimeter (DMM): Measures voltage, current, and resistance. Tests PSU output voltages, cable continuity, component values, and verifies circuit integrity. 4. Compressed Air Can: Blows accumulated dust out of fans, heatsinks, keyboards, and vents. Restores airflow to prevent overheating. Must not use mouth — moisture damages components. 5. LAN Cable Tester: Verifies all 8 wires in an Ethernet network cable are connected correctly and transmitting — used after making new cables or diagnosing network connectivity problems.

(b) **6 marks** State the THREE types of maintenance with an example of each.

Model Answer: 1. Preventive Maintenance: Done REGULARLY before failures occur. Example: cleaning dust from fans and heatsinks every 3-6 months, running antivirus updates weekly, performing monthly backups. Prevents problems from happening. 2. Corrective Maintenance: Done AFTER a failure has already occurred. Example: replacing a failed hard drive, reinstalling an operating system after corruption, replacing a burnt-out PSU. Most expensive type. 3. Predictive Maintenance: Monitoring system for early warning signs of impending failure using diagnostic tools. Example: reading HDD SMART data for reallocated sectors, monitoring CPU temperature trends for gradual increases indicating cooling degradation.

(c) **4 marks** Explain correct first aid procedure for a workshop burn injury.

Model Answer: 1. Do NOT put ice, butter, toothpaste, or any home remedy on the burn — these cause further damage and increase infection risk. 2. Immediately cool the burn with clean, cool running water for a MINIMUM of 10 minutes — this removes heat from tissues and reduces depth of injury. 3. Do not break any blisters that form — they protect the wound from infection. 4. Cover loosely with a clean, non-fluffy bandage or cloth. Seek medical attention for any significant burn, especially on hands, face, or if larger than a 50-cent coin.

Question 2

20

Answer all parts of this question.

(a) **12 marks** Explain the resistor colour code system and calculate the value of a resistor marked: Yellow-Violet-Orange-Gold.

Model Answer: The resistor colour code is an internationally standardised system (IEC 60062) using coloured bands painted on resistors to indicate their resistance value and tolerance. This avoids the need to print tiny numbers on small resistors. For a 4-band resistor: Band 1 = First significant digit Band 2 = Second significant digit Band 3 = Multiplier (value = 10 raised to that colour's digit) Band 4 = Tolerance (Gold=±5%, Silver=±10%) Colour values (mnemonic: BB ROY Great Britain Very Good Wife): Black=0, Brown=1, Red=2, Orange=3, Yellow=4, Green=5, Blue=6, Violet=7, Grey=8, White=9 Calculation for Yellow-Violet-Orange-Gold: Yellow = 4 (first digit) Violet = 7 (second digit) Orange = 3 → multiplier = $10^3 = 1,000$ Gold = ±5% tolerance Value = $47 \times 1,000 = 47,000\Omega = 47k\Omega \pm 5\%$ Acceptable range = $47,000 \times 0.95 = 44,650\Omega$ to $47,000 \times 1.05 = 49,350\Omega$

(b) **8 marks** Describe how to use a multimeter to test: (i) a cable for continuity, (ii) a battery voltage, (iii) a resistor value.

Model Answer: (i) Cable continuity test: Set DMM dial to CONTINUITY (beep symbol). Power off circuit. Insert black lead into COM, red into V/ Ω terminal. Touch one probe to each end of the wire being tested. A BEEP indicates the wire is intact and conducting. No beep = wire is broken somewhere. Test each wire in a cable individually. (ii) Battery voltage: Set DMM to DC Voltage (V $\rightarrow$). Set range slightly above expected voltage. Touch red probe to positive (+) terminal, black probe to negative (-). Read display: AA battery $\approx 1.5V$

(below 1.2V = discharged), 9V battery $\approx$ 9V (below 7V = replace), 12V car battery $\approx$ 12.6V fully charged (below 11.5V = discharged). (iii) Resistor value: Power OFF completely. Set DMM to Ω (resistance) range — start at 200k Ω if unsure of value. Insert black lead into COM, red into Ω terminal. Touch probes to each lead of the resistor. Read value on display. Compare to calculated colour code value. If within tolerance = good resistor. Significantly different = damaged or wrong resistor.

Question 3

20

Answer all parts of this question.

(a) 12 marks Compare impact and non-impact printers. Give THREE examples of each and state ONE advantage and ONE disadvantage of each type.

Model Answer: **IMPACT PRINTERS:** Definition: Form characters by physically striking a mechanism against an ink ribbon that contacts the paper. Examples: (1) Dot-matrix printer (pins strike ribbon), (2) Daisy wheel printer (rotating character wheel), (3) Line printer (prints entire line at once) Advantage: Can print through multiple layers of paper — ideal for multipart forms, receipts, invoices Disadvantage: Generally noisy due to striking mechanism — not suitable for quiet office environments **NON-IMPACT PRINTERS:** Definition: Form characters without physically striking the paper — no mechanical contact between print mechanism and paper. Examples: (1) Laser printer (toner+drum+heat), (2) Inkjet printer (sprays ink droplets), (3) Thermal printer (heat-sensitive paper) Advantage: Much quieter operation — higher print quality achievable — faster speeds for laser Disadvantage: Cannot print multipart forms — single copy only per print job Key comparison: Dot-matrix = cheap running cost, handles harsh environments, low quality. Laser = expensive hardware, cheapest per page at high volume, best quality. Inkjet = cheap hardware, expensive ink per page, good colour.

(b) 8 marks Describe the complete process of how a laser printer produces a printed page.

Model Answer: A laser printer (also called a page printer) processes and stores the ENTIRE page in memory before printing begins. 1. CHARGING: The charging corona wire gives the selenium-coated drum surface a uniform negative electrical charge across its entire surface. 2. EXPOSING: The laser beam scans across the drum. Wherever the laser hits, it removes the negative charge — creating the image of the document as areas of lost charge on the drum. 3. DEVELOPING: The drum passes through the toner cartridge. Toner (fine dry powder carrying a negative charge) is attracted to the exposed (uncharged) areas on the drum — the image now exists as a layer of toner on the drum. 4. TRANSFERRING: The drum rolls over the paper. A transfer corona wire gives the paper a strong positive charge that pulls the negatively charged toner from the drum surface onto the paper. 5. FUSING: The paper passes between heated rollers (fuser unit at approximately 180-200°C) and pressure rollers. The heat melts the toner powder and the pressure permanently bonds it into the paper fibres. 6. CLEANING: A cleaning blade removes any remaining toner from the drum. The drum is discharged by an erase lamp. The drum is ready for the next page.

Question 4

20

Your workplace has received several old computers, monitors, and phones that are no longer working and need to be disposed of.

(a) 8 marks Identify FOUR hazardous materials found in these devices and explain the health effects of each.

Model Answer: 1. LEAD (found in CRT monitor glass, solder joints, batteries): Lead is a cumulative neurotoxin. Long-term exposure damages the brain and nervous system — causing memory loss, reduced IQ, developmental problems. Children are especially vulnerable as lead disrupts brain development permanently. 2. MERCURY (found in LCD monitor backlights, switches): Mercury is a potent neurotoxin that bioaccumulates in the food chain. It damages kidneys and causes neurological problems. When mercury reaches water bodies, fish absorb it — humans who eat contaminated fish also accumulate mercury (Minamata disease). 3. CADMIUM (found in rechargeable NiCd batteries, some semiconductors): Cadmium is a known human carcinogen. It accumulates in kidneys causing progressive kidney disease. It also causes bone softening (Itai-itai disease) by interfering with calcium metabolism. 4. BROMINATED FLAME RETARDANTS (found in plastic circuit board casings and cables): When burned during informal recycling, these release toxic dioxins and furans. These compounds cause liver damage, hormone disruption (endocrine disruption), and are suspected carcinogens. They persist in the environment for decades.

(b) 6 marks Explain how this e-waste should be managed at institutional level.

Model Answer: 1. Document all items in the IT asset register — record serial numbers, model, condition, and planned disposal method for each device. 2. Securely wipe all data from functional devices using data wiping software (multiple overwrites) before disposal. For non-functional storage devices, physically destroy them (degauss, shred, or drill through) to prevent data recovery. 3. Contact a certified e-waste recycling company — verify they have proper licences and facilities for hazardous electronic waste. Obtain certificates of disposal for all items. 4. Separate working equipment from non-working — donate functional items to schools or charities first (extending useful life). Only truly dead equipment goes to recyclers. 5. Record the disposal in the asset register — document which items went where, date, and recycler name — for auditing and legal compliance purposes.

(c) 6 marks Describe the e-waste management cycle from collection to final disposal.

Model Answer: 1. COLLECTION: Designated e-waste collection points are established at offices, schools, retailers, and public locations. Communities are educated to bring old electronics here instead of dumping in general waste. 2. SORTING: Collected waste is categorised — IT equipment, consumer electronics, batteries, mercury-containing lamps — separated because each stream requires different specialist processing. 3. DISASSEMBLY: Items are carefully taken apart by trained workers to separate: plastic housings, metal frames, circuit boards, cables, screens, and batteries. Manual disassembly recovers more materials than shredding. 4. MATERIAL RECOVERY: Valuable materials are extracted — precious metals (gold, silver, palladium) from circuit boards, copper from wiring, aluminium from casings, plastics sorted for recycling. This generates revenue that partially funds the recycling operation. 5. SAFE DISPOSAL: Remaining hazardous materials — lead from CRTs, mercury from lamps, cadmium from batteries, battery acid — are processed through specialist licensed hazardous waste facilities. These use controlled processes to neutralise, immobilise, or safely contain toxins preventing environmental release.

Question 5

20

Answer all parts of this question.

(a) 8 marks Explain the THREE types of UPS and recommend the most appropriate one for Uganda. Give detailed reasons.

Model Answer: 1. OFFLINE/STANDBY UPS: Monitors the mains supply. When mains fails, it switches to battery. There is a brief switchover delay (typically 5-25 milliseconds). No voltage regulation — does not protect against voltage fluctuations, surges, or brownouts. Cheapest type. Only protects against complete power loss. 2. LINE-INTERACTIVE UPS: Contains a voltage regulator (AVR — Automatic Voltage Regulator) that handles brownouts and surges by adjusting voltage WITHOUT switching to battery. Only switches to battery during complete power failure. Switchover is fast (< 5ms). Protects against both voltage instability AND outages. Moderate cost. 3. ONLINE/DOUBLE-CONVERSION UPS: Power always flows through the battery — mains charges battery, battery continuously powers equipment. Zero switchover time (no gap ever). Provides the cleanest, most stable power output regardless of mains quality. Most expensive. Used for servers and critical medical equipment. RECOMMENDATION FOR UGANDA: Line-interactive UPS. REASONS: Uganda experiences BOTH types of power problems — (a) frequent voltage fluctuations, brownouts, and surges from the grid, AND (b) actual complete power outages. The offline/standby UPS only handles complete outages and does NOT regulate voltage, leaving equipment vulnerable to fluctuations. The online UPS would be ideal but is prohibitively expensive for most organisations. The line-interactive UPS handles both problems at a reasonable cost — it is the practical choice for Uganda.

(b) 6 marks Describe how to connect a computer system correctly. List the steps in order.

Model Answer: 1. Place the monitor at a comfortable eye level — centre of screen should be at or slightly below eye level, approximately 50-70cm from the user to reduce eye strain. 2. Connect the video cable (VGA, HDMI, or DisplayPort) firmly from the back of the monitor to the corresponding port on the PC graphics card or motherboard. 3. Connect keyboard to a USB port (or PS/2 port on older systems) at the back of the PC. Connect mouse to another USB port. 4. Connect the network cable (RJ-45 Ethernet) from the wall port or switch to the NIC port at the back of the PC. 5. Connect power cables — monitor power to wall socket, PC power cable to wall socket. IMPORTANT: Verify PSU voltage selector is set to 220-240V (Uganda mains voltage) before connecting. 6. Power on the MONITOR first, then press the PC power button. This sequence ensures the monitor is ready to display POST output.

(c) 6 marks State FIVE energy-saving configurations for computers (Energy Star).

Model Answer: 1. Enable sleep/standby mode after a period of inactivity (typically 10-20 minutes) — computer reduces power consumption significantly to around 3-10W while sleeping instead of 100-300W running. 2. Set monitor to turn off automatically after 10-15 minutes of no activity — monitor is often the biggest power consumer and can draw 20-80W. 3. Enable hibernate after extended inactivity — saves complete RAM contents to disk and powers off — uses near zero power but can resume exactly where left off. 4. Enable power management features in BIOS/UEFI — allows OS to control CPU power states (C-states and P-states) — CPU runs at lower speed/voltage when not fully loaded. 5. Unplug all peripheral devices not currently in use (printers, external drives, USB hubs) — devices in standby still draw phantom/vampire power that adds up over time across many devices.

Quick Reference — Key Facts for the Exam

Assessment Breakdown

Component	Weight
Test	10%
Assignment	10%
Practical (Lab work)	20%
Final Examination	60%
TOTAL	100% — Pass: 50%

Unit 1 — Tools & Safety

- 6 tool categories: Hand tools, ESD, Diagnostic, Cleaning, Networking, Support equipment
- ESD strap = MOST IMPORTANT tool — always worn during internal work
- CO2 extinguisher ONLY for electrical/computer fires — NEVER water or foam
- Electric shock: DISCONNECT POWER FIRST before touching victim
- Never work alone in the workshop

Unit 2 — System Usage & Maintenance

- 3 maintenance types: Preventive (before failure), Corrective (after failure), Predictive (monitor signs)
- Defragment HDD only — NEVER defragment SSD
- 3-2-1 backup rule: 3 copies, 2 media types, 1 offsite
- Energy Star: sleep mode, monitor off, BIOS power management, unplug unused devices

Unit 3 — Electronics & Multimeter

- Passive: Resistor, Capacitor, Inductor, Transformer, Fuse
- Active: Diode, Transistor, Integrated Circuit (IC)
- Mnemonic: BB ROY Great Britain Very Good Wife = Black Brown Red Orange Yellow Green Blue Violet Grey White = 0-9
- Gold= $\pm 5\%$, Silver= $\pm 10\%$
- 4-band: Digit1 – Digit2 – Multiplier – Tolerance
- DMM: Power OFF for resistance/continuity. Meter in SERIES for current.
- Continuity: beep = complete. No beep = open/broken.

Unit 4 — Printers & Scanners

- Impact: Daisy wheel, Dot-matrix, Line printer. Non-impact: Inkjet, Laser, Thermal, 3D, Plotter
- ONLY dot-matrix can print multipart forms
- Laser: Charge → Expose → Develop → Transfer → FUSE → Clean (6 steps)

- Minimum scan DPI: 300 (documents), 600+ (photos)
- PRE-SCAN always first — low-res preview before final scan

Unit 5 — Power Packs & UPS

- Wire colours: Yellow=+12V, Red=+5V, Orange=+3.3V, Black=GND, Green=PS_ON
- UPS: Offline (cheapest), Line-interactive (Uganda best), Online (most expensive)
- Paperclip test: short GREEN to BLACK → fan spins = PSU alive
- +12V range: 11.4V-12.6V. +5V: 4.75V-5.25V. +3.3V: 3.13V-3.46V

Unit 6 — E-Waste

- Lead (CRT/solder) → brain/nervous system. Mercury (LCD) → food chain/kidneys. Cadmium (batteries) → cancer/kidneys
- 3Rs in order: Reduce → Reuse → Recycle
- E-waste cycle: Collection → Sorting → Disassembly → Recovery → Safe Disposal
- Management: Personal, Institutional, Government levels
- NEVER burn e-waste — releases toxic dioxins